

GenAI Expert Chatbot

Project Idea:

A chatbot specialized in Generative AI that answers questions about topics such as Transformers, GPT models, GANs, VAEs, Diffusion models, RAG, fine-tuning, prompt engineering, and MLOps. The chatbot provides technically accurate, detailed, and structured explanations, acting as a virtual tutor for AI learners and researchers.

Github repo URL:

[Link](#)

Tools and libraries

- Python (for data processing, model fine-tuning, inference)
- PyTorch (for LLM fine-tuning and GAN training)
- HuggingFace Transformers (pre-trained LLM and tokenizer)
- FAISS / Chroma (for vector database for RAG retrieval)
- NLTK / SpaCy (text preprocessing)
- FastAPI (for chatbot deployment interface)
- MLflow (for model tracking, versioning, and MLOps)
- Google Colab (for model training)

Project Team Members

- Mostafa Mahmoud Mohamed Noseir
- Mohamed Mahmoud Mohamed Noseir
- Mohamed Abdul-Shafi Siddiq
- Nourseen Khaled Joudah Ibrahim
- Rana Sameh Awad Abdalla
- Fatma Mohamed Ahmed Othman

Team Leader

- Mostafa

Final Project Deliverable

A fully functional **AI-powered chatbot** deployed as a **web app / mobile app**. The system integrates a fine-tuned Large Language model with a RAG-based retrieval system and includes an end-to-end automated pipeline with MLOps tracking.

Machine Learning Pipeline

1. Data Collection

- Raw GenAI documents (blogs, books, papers)
- Q&A pairs for supervised fine-tuning



2. Data Preprocessing

- Cleaning, deduplication, keyword filtering for GenAI relevance
- Tokenization, normalization (lowercasing, stopword, lemmatization)
- Splitting into train/validation sets



3. Model Training

- Fine-tune a modern small instruct LLM on Q&A dataset
- GAN to generate creative summaries or suggested study topics



4. RAG System

- Embed curated documents into vector database
- Retrieve relevant context based on query similarity



5. Inference

- Combine retrieved context + user query into prompt
- Generate structured answer with the fine-tuned LLM



6. Post-processing

- Formatting, Coherence/quality check



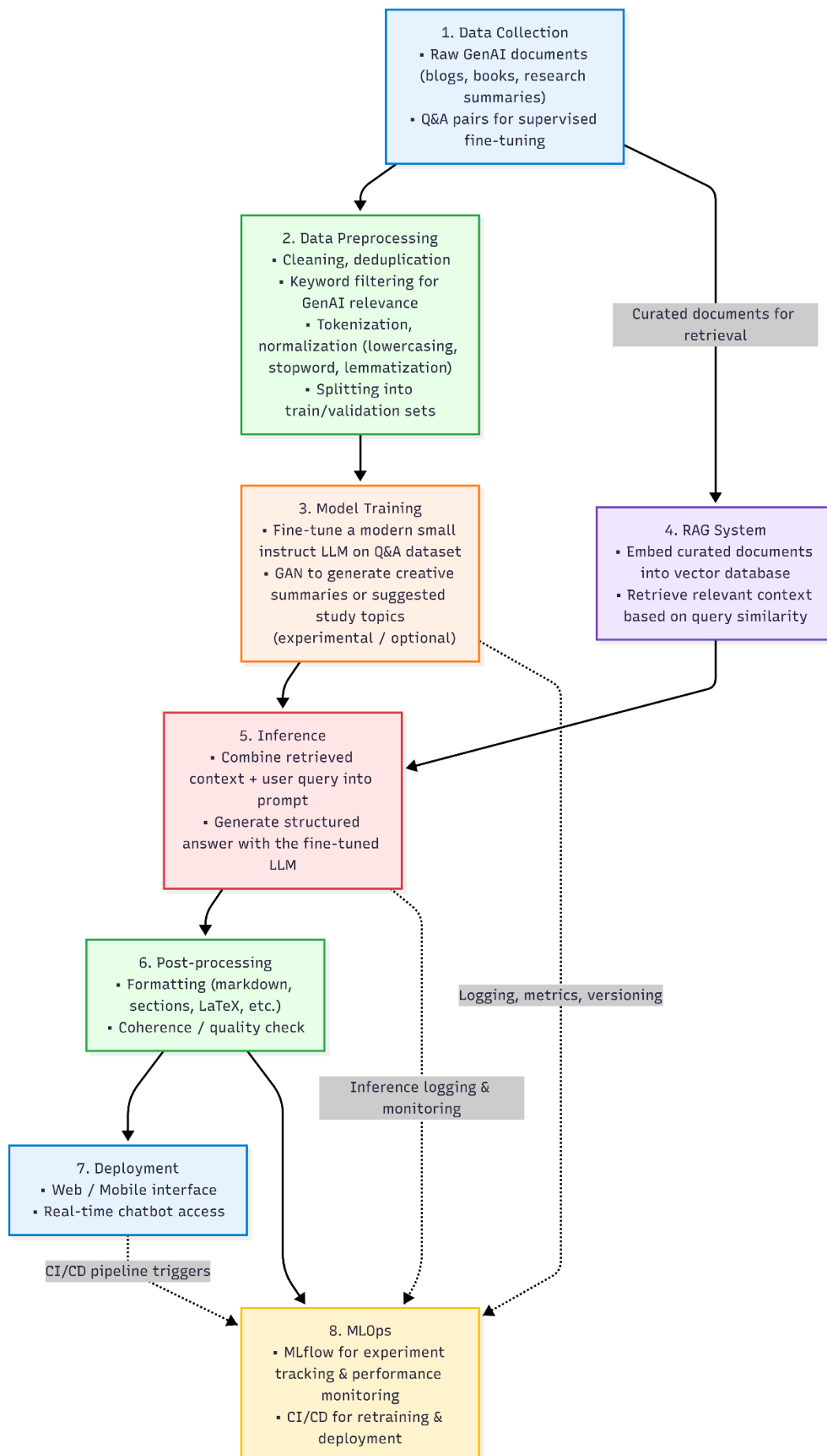
7. Deployment

- Web / Mobile interface



8. MLOps

- MLflow for experiment tracking & performance monitoring
- CI/CD for retraining/deployment



Dataset source

- Kaggle NLP / LLM papers:
- HuggingFace Q&A datasets
- ArXiv AI papers
- Stanford Question Answering Dataset

Timeline Deliverables

Week 1 & 2

Weeks deliverables:

- Collect GenAI Papers and Books

Team Members' Roles:

- Mostafa & Mohamed: Data Collection Papers
- Rana & Nourseen: Data Collection Books

Week 3 & 4

Weeks deliverables:

- Data Extraction & Format Conversion (convert all raw data into markdown)

Team Members' Roles:

- Mostafa & Mohamed & Rana & Nourseen: Data Extraction

Week 5 & 6

We are in holiday

Week 7 & 8

Weeks deliverables:

- Data Cleaning
- Chunking
- Add metadata is attached to each chunk

Team Members' Roles:

- Rana & Nourseen: Data Cleaning
- Mostafa & Mohamed: Chunking & Add metadata is attached to each chunk

Week 9 & 10

Weeks deliverables:

- Create vector embeddings of documents for RAG
- **RAG System Validation:** Test the retrieval component to ensure it provides accurate and relevant context based on query similarity

Team Members' Roles:

- Mostafa & Mohamed: Create vector embeddings of documents for RAG
- Rana & Nourseen: **RAG System Validation:** Test the retrieval component to ensure it provides accurate and relevant context based on query similarity